

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 5: Belt Transect Across a Changing Habitat: How Does Plant Population Change as Conditions Change?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
C	A	B	C	C	B	A	B

USE AFTER: PREDICT PHASE — Predict: sketching a prediction of plant abundance across the four zones and writing one sentence explaining why

Q1. Before collecting data, a learner predicts that a shaded area of the school compound will have fewer grass plants per square metre than the open field. Which reason best supports this prediction?

- A. Plants in the shade do not need any water
- B. Grass grows best when it gets no sunlight at all
- C. Shade gives less sunlight, so the grass can make less food by photosynthesis**
- D. Shaded areas always have more trampling than open fields

Correct answer: C. Light is an abiotic factor that limits plant growth. Less light means less photosynthesis, so fewer plants are expected in the shade.

USE AFTER: OBSERVE PHASE — Observe: laying out the belt transect and placing quadrats every 2 m between the two lines

Q2. In a belt transect study, a group places quadrats at regular intervals (every 2 m) between two parallel lines instead of choosing the greenest spots. Why is this done?

- A. It avoids bias, so the sample represents the whole transect and not only the best places**
- B. It makes sure every quadrat contains the same number of plants
- C. It protects the plants from being trampled by the learners
- D. It makes the counting finish faster because there are fewer plants to count

Correct answer: A. Regular spacing removes personal choice. This gives a fair picture of how plant numbers change along the whole transect.

USE AFTER: EXPLAIN PHASE — Explain: calculating the mean abundance for each zone in home groups

Q3. Team A counted the plants in five 1 m x 1 m quadrats placed in the open field: 16, 20, 18, 22 and 14 plants. What is the mean abundance per quadrat in the open field?

- A. 22 plants
- B. 18 plants**
- C. 90 plants
- D. 20 plants

Correct answer: B. Mean = total plants ÷ number of quadrats = $(16 + 20 + 18 + 22 + 14) \div 5 = 90 \div 5 = 18$ plants per quadrat.

USE AFTER: EXPLAIN PHASE — Explain: estimating density (plants per square metre) from the quadrat data

Q4. A group uses a 50 cm x 50 cm quadrat (area 0.25 m²) in the garden zone. The mean count is 3 plants per quadrat. What is the estimated density in plants per square metre?

- A. 0.75 plants per m²
- B. 6 plants per m²
- C. 12 plants per m²**
- D. 3 plants per m²

Correct answer: C. Density = mean count ÷ area of quadrat = $3 \div 0.25 = 12$ plants per m². Four of these small quadrats make 1 m².

USE AFTER: EXPLAIN PHASE — Explain: writing a claim-evidence-reasoning explanation using the light, moisture and trampling data

Q5. The school pathway gets plenty of sunlight, but the transect showed very few plants growing on it. What is the best explanation?

- A. Plants on pathways have no competition, so they cannot grow
- B. Too much sunlight kills all plants
- C. Constant trampling damages plants and compacts the soil, so few plants can grow**
- D. Pathways have no soil at all

Correct answer: C. Trampling is a limiting factor. It crushes plants and compacts the soil even where light is plentiful.

USE AFTER: EXPLAIN PHASE — Explain: comparing school compound factors with river factors (nutrients from farms and sewage)

Q6. In the school garden, manure and water were added and weeds grew very thickly. A river near a farm receives fertiliser and sewage, and algae and aquatic weeds grow thickly there too. Which statement best links the two situations?

- A. Fish eat the algae, so the algae are the cause of the fish disappearing
- B. Extra nutrients boost the growth of plants and algae in both places, as the manure does in the garden**
- C. Extra nutrients reduce the growth of algae in rivers but increase it in gardens
- D. Algae and weeds in the river are many, so the river must be healthier now

Correct answer: B. Added nutrients (manure or fertiliser and sewage) remove a limit on growth, so plants or algae increase. More plants does not always mean a healthier habitat.

USE AFTER: MODEL BUILDING PHASE — Model: adding abiotic and biotic factors affecting algae, weeds and fish to the river model

Q7. Which pair of factors that could affect fish in a river contains one abiotic factor and one biotic factor?

- A. Oxygen level in the water and competition for food among fish**
- B. Algae and aquatic weeds
- C. Water temperature and nutrient level
- D. Sunlight and water temperature

Correct answer: A. Oxygen level is a non-living (abiotic) factor, and competition for food between living organisms is a biotic factor. The other pairs are both abiotic or both biotic.

USE AFTER: END OF LESSON — End of lesson: exit ticket on factors affecting plant population size

Q8. In the shaded part of the school compound, few plants grew because there was little light. In ecology, what do we call a condition like this that restricts how large a population can grow?

- A. A quadrat
- B. A limiting factor**
- C. A habitat
- D. A niche

Correct answer: B. A limiting factor, such as light, nutrients, space or trampling, restricts population size. A niche is the role of an organism, a habitat is where it lives, and a quadrat is a sampling frame.